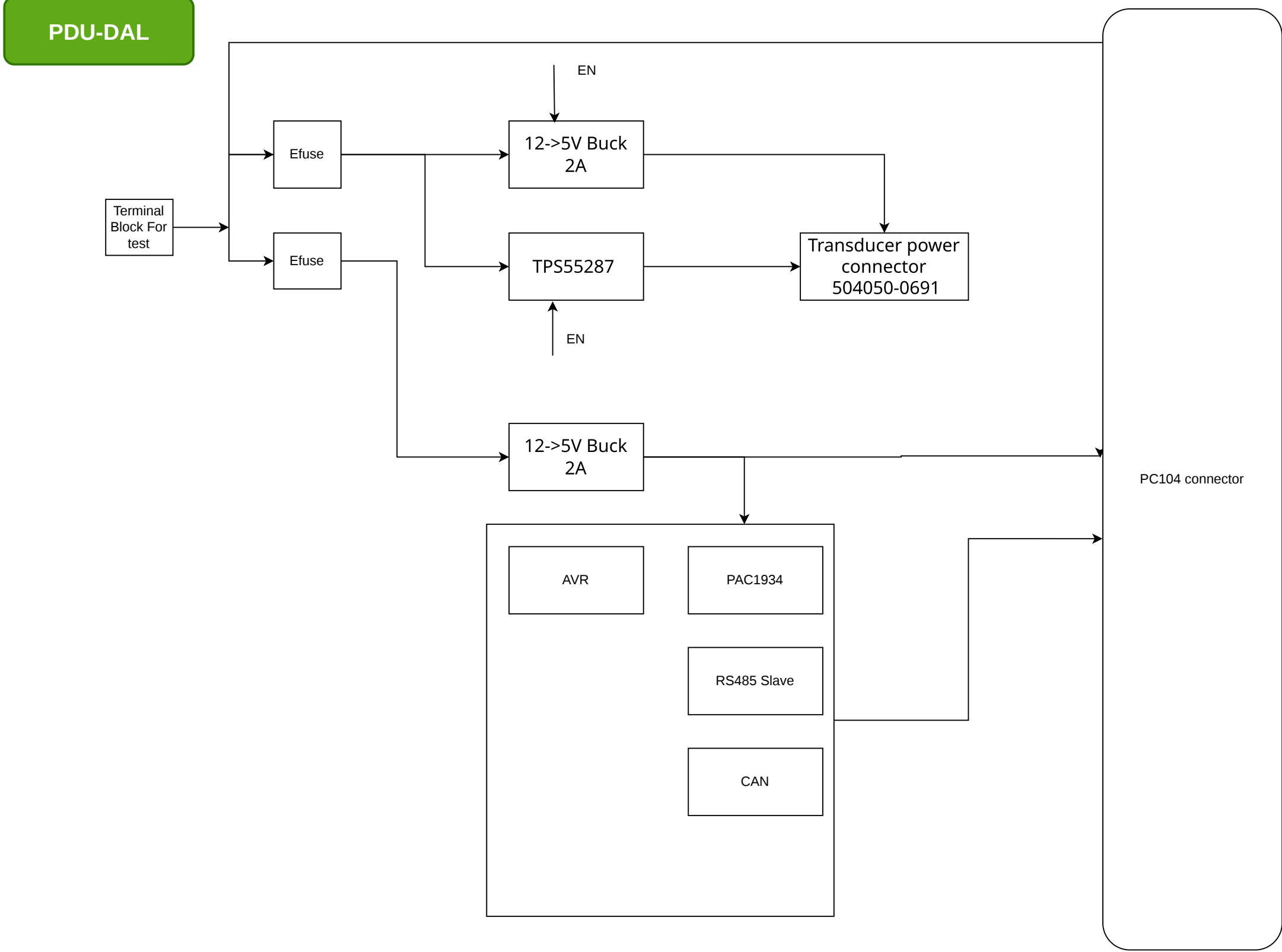
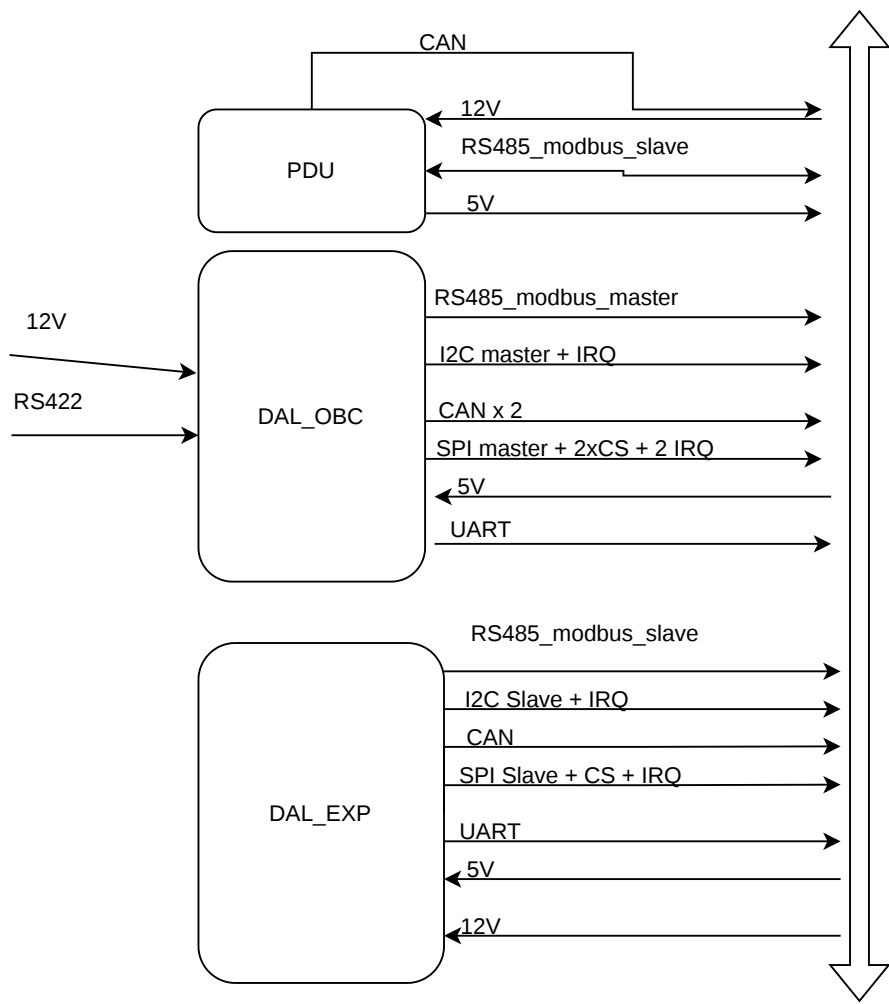




CAN_L*	1		
CAN_H*	3		
	5		
	7		
RS-422 RX+	9		
RS-422 RX-	11		
RS-422 TX+	13		
RS-422 TX-	15		
	17		
OBC_UART_A_TX	19	20	OBC_UART_A_TX
SPI CS1	21		SPI CS2
SPI SCLK	23		SPI IRQ1
SPI MOSI	25		SPI IRQ2
SPI MISO	27		
	29		
	31		
	33		
	35	36	
RS-485 B (-)	37	38	RS-485 A (+)
OBC_UART_C_RXD	39		
OBC_UART_C_TXD	41		
I2C_IRQ1	43	44	I2C2 SDA
I2C_IRQ2	45	46	I2C2 SDL
	47		
	49		
	51		

	1		
	3		
	5		
	7		
	9		
	11		
	13		
	15		
	17		
	19		
	21		
STX 12V	23		STX 12V
BUS 5V	25		BUS 5V
	27		
GND	29		GND
GND	31		GND
	33		
	35	36	
	37		
	39		
	41		
	43		
	45		
GND	47		GND
	49		
	51		

4.1.1 HI (PC104)	
Description	HI
CAN0_L	
CAN0_H	
GPIO_PC4(TR)	
-	
RS422_RX1+	
RS422_RX1-	
RS422_TX1+	
RS422_TX1-	
-	
UART0_RX	
SPI_CS0	
SPI_SPCK	
SPI_MOSI	
SPI_MISO	
-	
-	
-	
RS-485 #1 B	
OEM719_UART2_TXD	
I2C1_SDA	
I2C1_SCL	

H1 (ESQ-126-14-G-D)		Description
1	2	-
3	4	-
5	6	GPIO PC3 (STX_EN)
7	8	GPIO PC5 (nFPGA RESET)
9	10	GPIO PA29 (ENABLE)
11	12	GPIO PA28 (BOOT)
13	14	PPS OEM719
15	16	-
17	18	-
19	20	UART0_TX
21	22	-
23	24	-
25	26	-
27	28	-
29	30	-
31	32	-
33	34	-
35	36	-
37	38	RS-485 #1 A
39	40	I2C0 SDA
41	42	I2C0 SCL
43	44	I2C2 SDA
45	46	I2C2 SCL

HIỆN CÓ

I2C0	flow sensor
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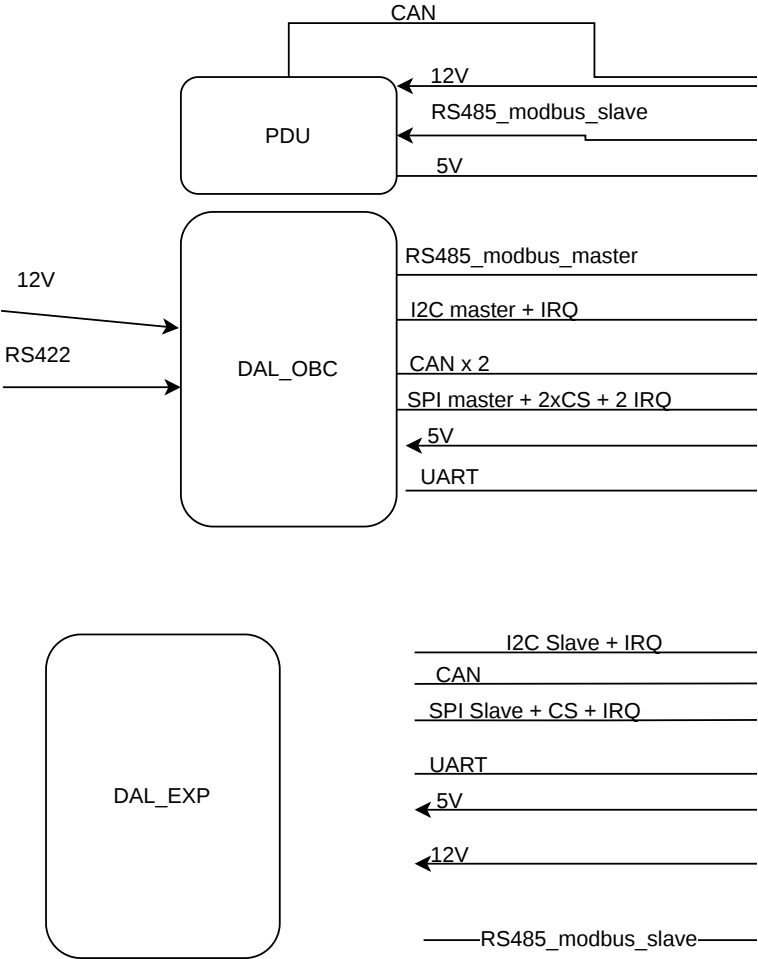
CẦN THÊM

I2C	CAM	
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- ĐO 5V ĐI VÀO
- ĐO NGUỒN LED IF
- ĐO NGUỒN STEP MOTOR

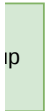
- CẦN RA HEADER CHO 5V (backu

PC 104 Signal

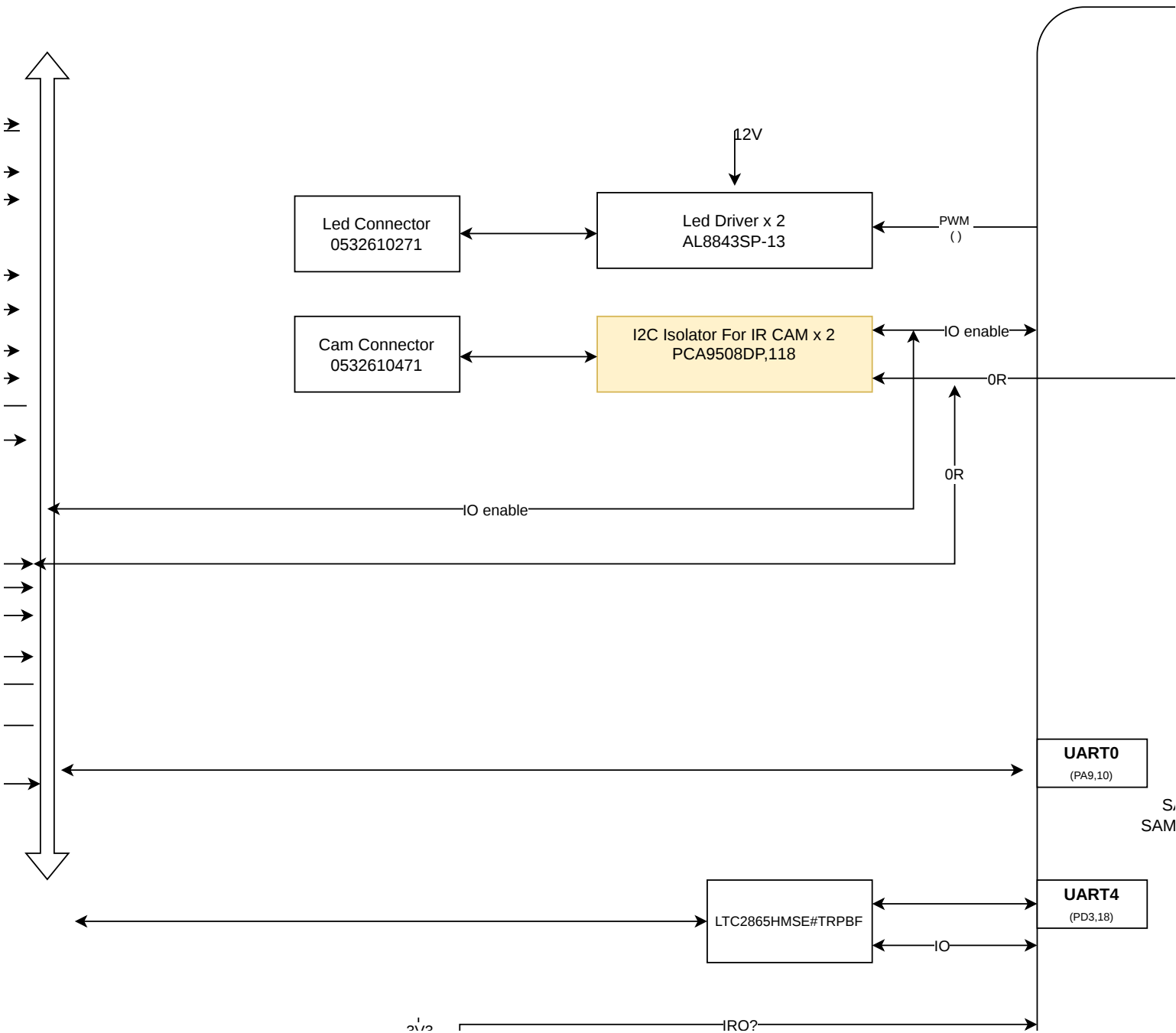


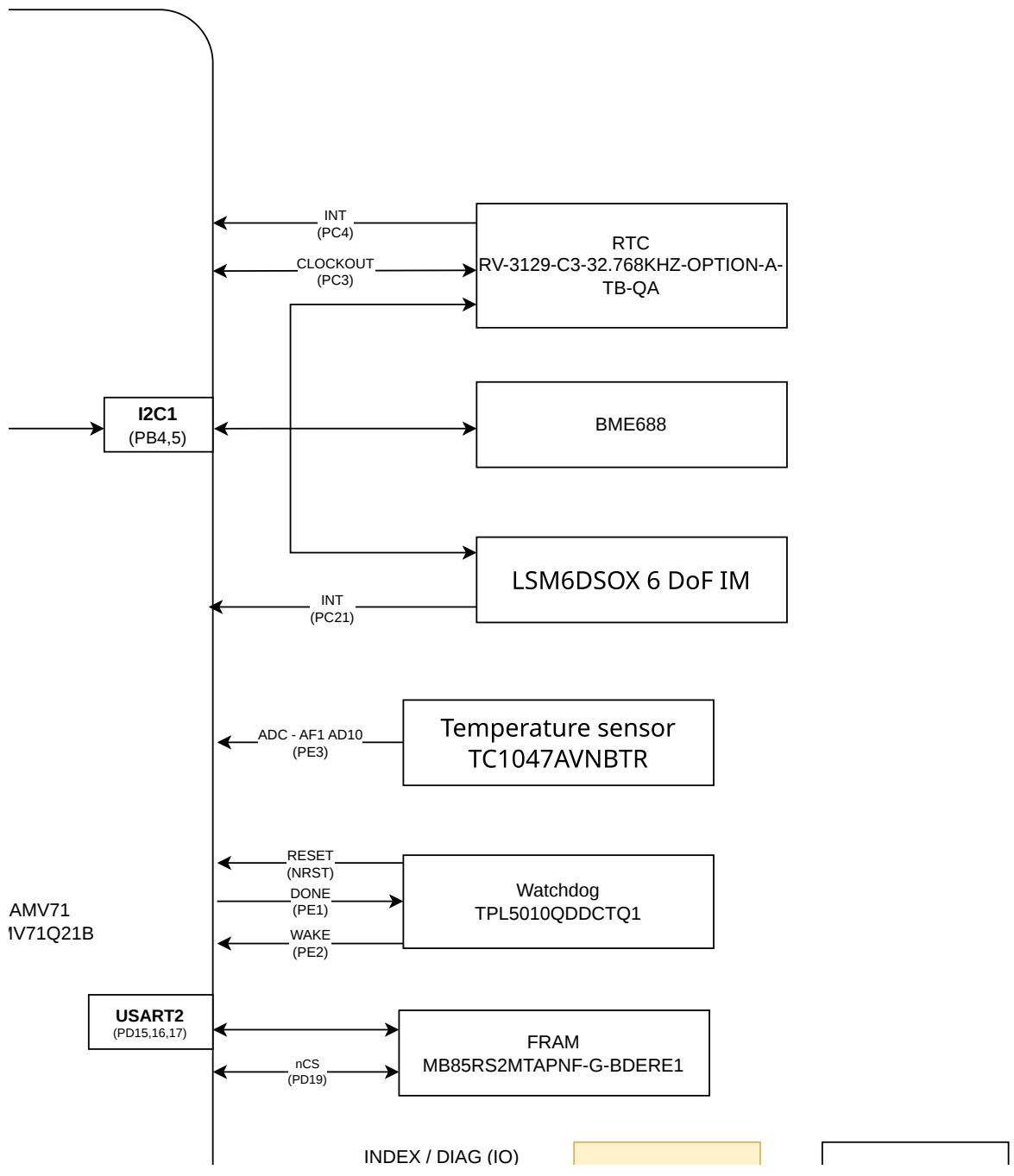
NOTE

đang hết i2c	SIGNAL	CHANEL
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EXP





CAN_L*	1		
CAN_H*	3		
	5		
	7		
RS-422 RX+	9		
RS-422 RX-	11		
RS-422 TX+	13		
RS-422 TX-	15		
	17		
OBC_UART_A_TX	19	20	OBC_UART_A_TX
SPI CS1	21		SPI CS2
SPI SCLK	23		SPI IRQ1
SPI MOSI	25		SPI IRQ2
SPI MISO	27		
	29		
	31		
	33		

	1	
	3	
	5	
	7	
	9	
	11	
	13	
	15	
	17	
	19	
	21	
STX 12V	23	
BUS 5V	25	
	27	
GND	29	
GND	31	
	33	

	STX 12V
	BUS 5V
	GND
	GND

	power sense
	HD PUMP
I2C1	RTC
	BME688
	LSM6DSOX
	CAM
I2C2	PC104
USART2	SOLENOID
	FRAM
UART2	CONSOLE

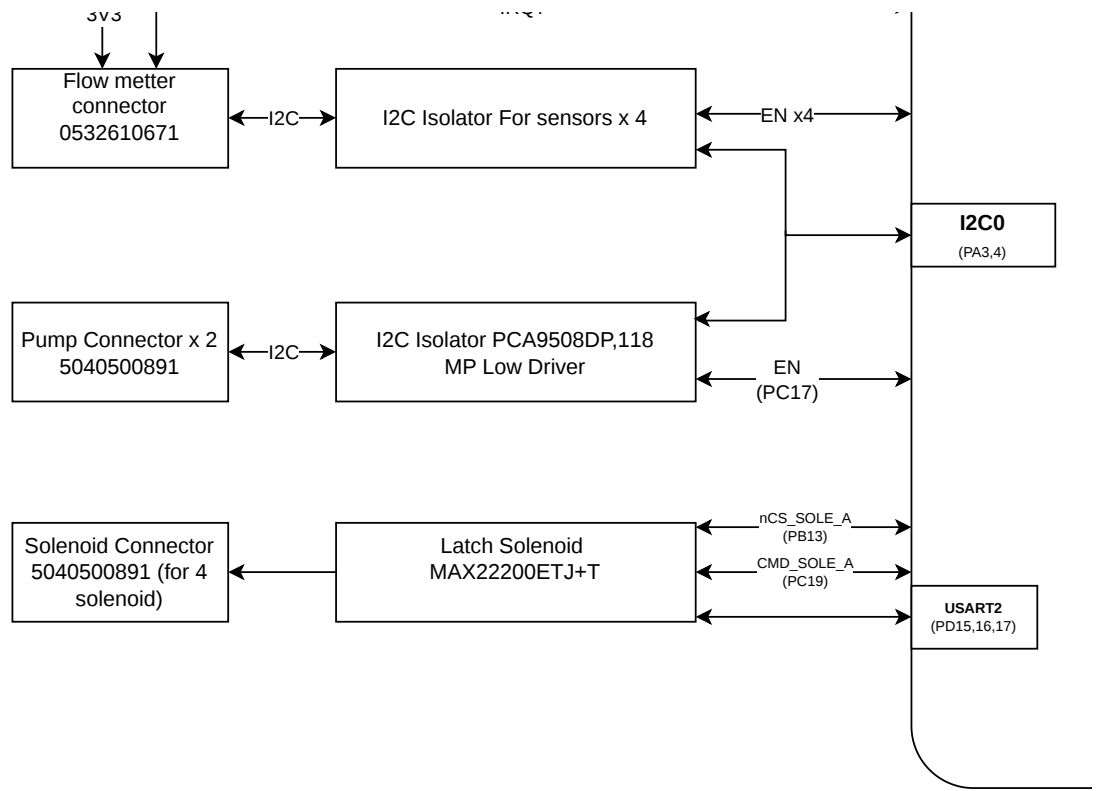
	ra bus PC104	
UART	STEP MOTOR x3	
	RS458	
	ra bus PC104	
PWM	STEP MOTOR x3	dùng

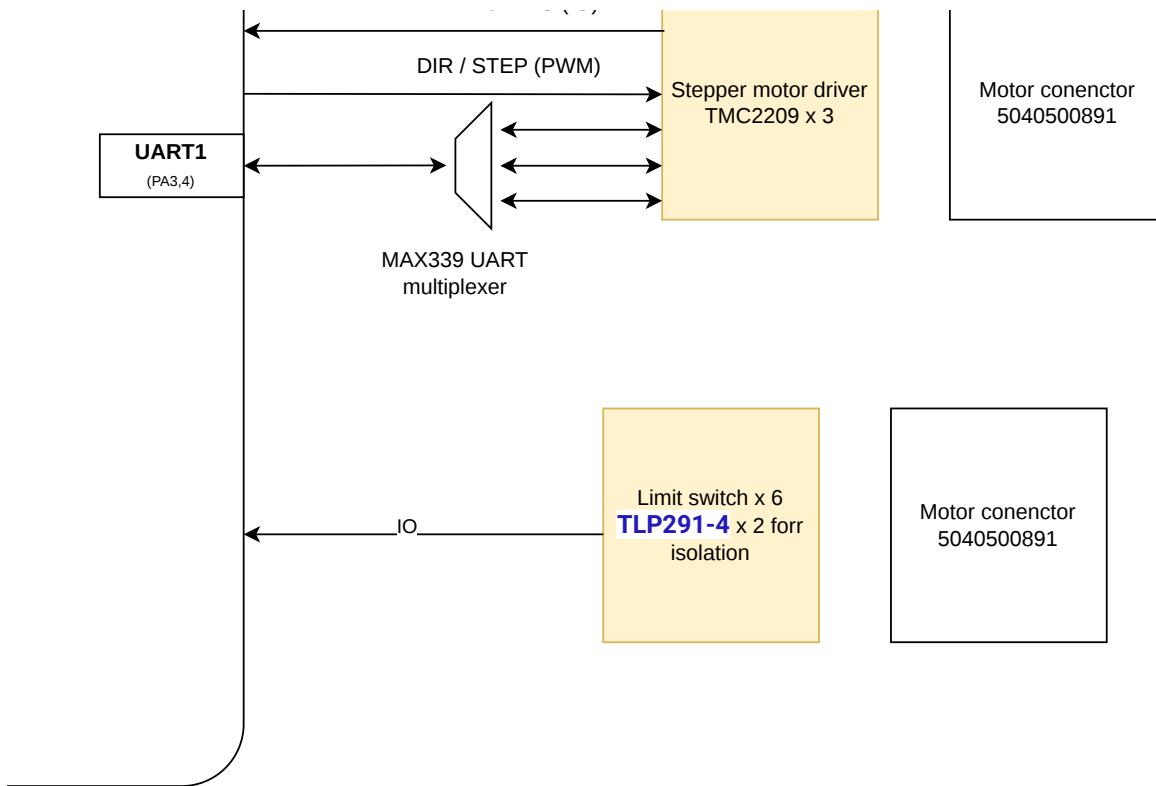
CÒN TRỐNG

PERIPHERALS	PIN	Note
UART0	- RXD0 (PB0), TXD0 (PB1) - RXD0 (PA9), TXD0 (PA10)	- EXP V10(BEE-PC2) dùng RXD0 (PA9), TXD0 (PA10) đến conn OBC - EXP V110(BEE-1012) dùng RXD0 (PA9), TXD0 (PA10) đến MPU
CAN0	CANTX0 (PB2), CANRX0 (PB3)	

1 I2C RIÊNG	I2C	3 ([0:2])
UART1		
UART4		
UART0		
ng PA27, PA0, PA1 (dùng từ chân heater cũ)		

DO
XD0





	33		
	35	36	
RS-485 B (-)	37	38	RS-485 A (+)
OBC_UART_C_RXD	39		
OBC_UART_C_TXD	41		
I2C_IRQ1	43	44	I2C2 SDA
I2C_IRQ2	45	46	I2C2 SDL
	47		
	49		
	51		

	33	
	35	
	37	
	39	
	41	
	43	
	45	
GND	47	
	49	
	51	

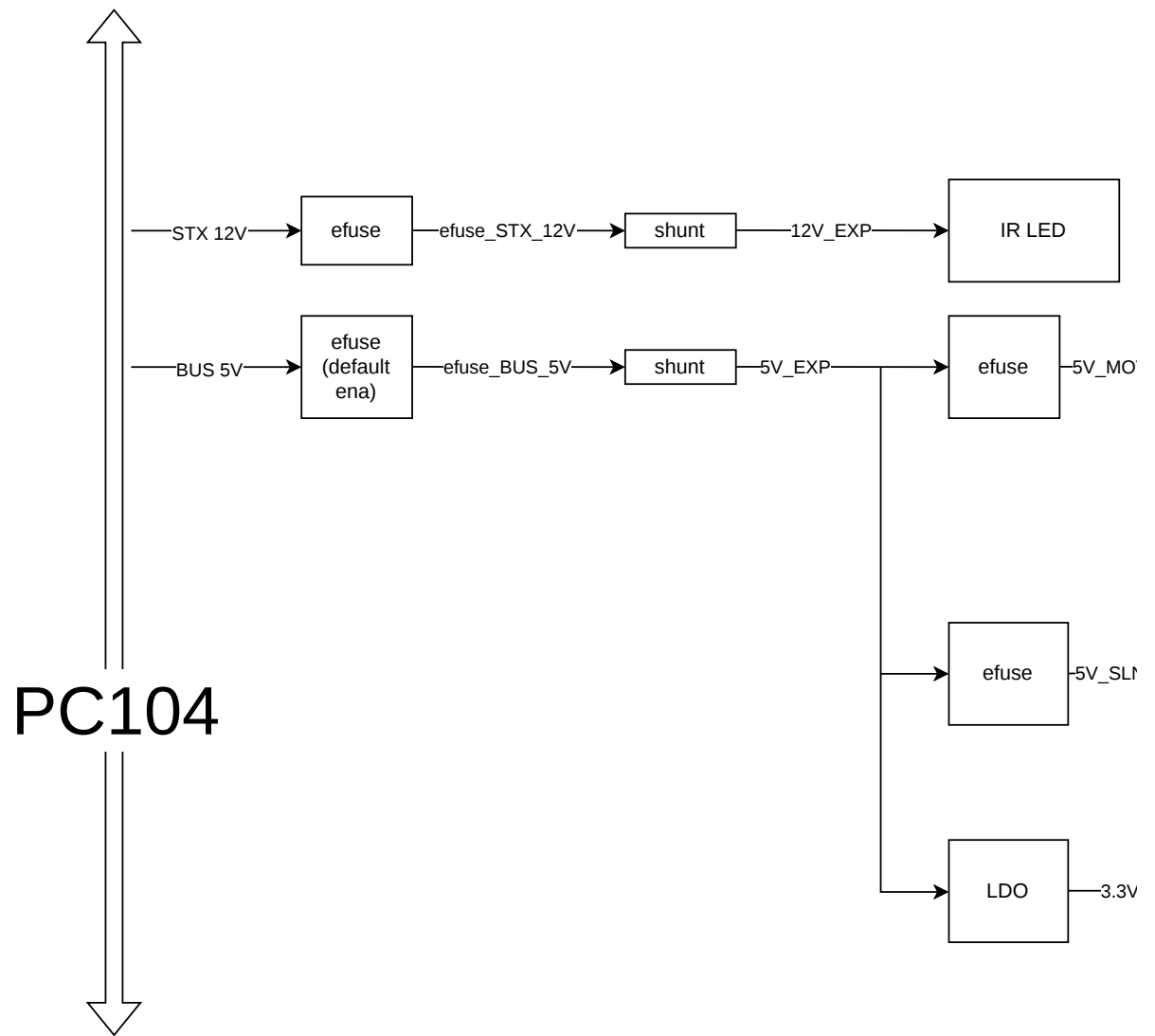
36	
	GND

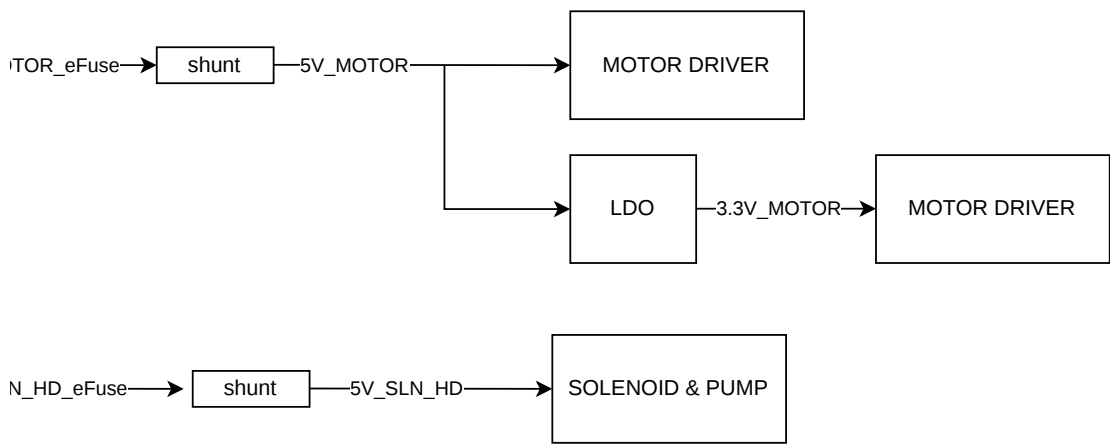
OLD (30/6/2025)

CAN_L*	1		
CAN_H*	3		
	5		
	7		
RS-422 RX+	9		
RS-422 RX-	11		
RS-422 TX+	13		
RS-422 TX-	15		
SPI CS2	17		SPI IRQ2
SPI CS1	21		SPI IRQ1
SPI SCLK	23		
SPI MOSI	25		
SPI MISO	27		
	29		
	31		
	33		

STX :
BUS
GN
GN

	1		
	3		
	5		
	7		
	9		
	11		
	13		
	15		
	17		
	21		
12V	23		STX 12V
5V	25		BUS 5V
	27		
D	29		GND
D	31		GND
	33		





`/_MCU` →

		35			
	RS-485 B (-)	37	38	RS-485 A (+)	
	UART_TXD	39		UART_RXD	
		41			
	I2C_IRQ1	43		I2C2 SDA	
	I2C_IRQ2	45		I2C2 SDL	
		47			GN
		49			
		51			

	35			
	37	38		
	39			
	41			
	43			
	45			
D	47		GND	
	49			
	51			

